MIL-T-36310(DM)
21 October 1964

MILITARY SPECIFICATION

TURNING FRAME, ORTHOPEDIC BED,
STRYKER, 84 INCH FRAME

1. SCOPE

1.1 This specification covers one type of orthopedic turning frame, for use (see 6.1) in orthopedic, neurosurgical, and gynecological services where immobilization of the spine, with or without hyperextension, is required, and where frequent turning of the patient is necessary for proper nursing care.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

UU-P-271	Paper, Wrapping, Waterproofed Kraft.
CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.
CCC-C-430	Cloth, Cotton, Sheeting.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple, Wall.

6530

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

Fed. Std. No. 66 Steel: Chemical Composition and Hardenability.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

Defense Medical Supply Center

20260 (2 sheets)	Turning Frame, Orthopedic Bed, Stryker, 84 inch Frame.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. All materials used in fabrication of the orthopedic turning frame shall be as specified herein. Commercial tolerances on gage sizes shall apply.

3.1.1 Steel. The tubular supporting runners, the anterior and posterior frames, and the overhead frame shall conform to Fed. Std. No. 66 for C1010 carbon steel, or AISI 302 or 304 corrosion-resisting steel.

3.1.2 Bleached sheeting cotton. Cotton sheeting shall conform to CCC-C-430 for type II, class 2, style A.

3.1.3 Canvas. All canvas components shall be made from unbleached cotton, duck, conforming to CCC-C-419 for type I, duck number 11.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

3.2 Design. The orthopedic turning frame shall be as shown on DMSC Drawing 20260. Means shall be provided whereby the supporting runners can be securely bolted to the orthopedic bed cart (see 6.1).

3.3 Construction. The orthopedic turning frame shall consist of supporting runners, an anterior frame, a posterior frame, and an overhead frame. The parts used in fabrication of turning frame shall be as indicated in table I. The anterior and posterior frames shall be attached to the supporting runners by means of turning discs so that they can be fully rotated and locked into position by means of locking pins, in order to reverse a patient's position on the frame. The turning discs shall provide for utilization of the Crutchfield-type traction and counter-traction by means of a centrally located aperture. The overhead frame shall be attached to the supporting runners by means of a Y-connection, and shall be held securely by means of thumbscrews. The distance between the anterior and posterior frames shall be adjustable.

TABLE I. LIST OF PARTS

Part No.	Req'd.	Description
1	1	Overhead frame
2	4	Knurled nut, 1-1/4 by5/8 - 13
3	2	Turning disc, complete
4	1	Face support assembly
-	1	Face support canvas
-	1	Quilted face pad
5	1	Posterior frame
-	1	Posterior canvas cover set
-	1	Posterior mattress, cotton
-	2	Posterior mattress, end sections, foam rubber
-	1	Posterior mattress, center section, foam rubber
-	1	Posterior sheet
6	4	Stud bolt, 3-1/2 by 1/2
7	4	Knurled nut, 1-1/4 by 5/16 - 1/2 - 13
8	3	Thumb screw, 1/4 - 20 by 1/2
9	2	Walnut panel
10	2	Locking pin nut
11	1	Foot support frame
-	1	Foot support canvas
12	2	"Y" turning, complete
13	4	Set screws, 1/4 - 20 by 3/8
14	1	Anterior frame

Source: https://assist.dla.mil -- Downloaded: 20260-21T08:23Z
Check the source to verify that this is the current version before use.

TABLE I.

Part No.	Req'd.	Description
-	1	Anterior canvas cover
-	1	Anterior mattress, cotton
-	1	Anterior mattress, foam rubber
--	1	Anterior sheet
15	3	End swivel, long
16	8	Bushing, 1/2" - 18 gage by 1" long
17	2	Support runners
18	1	Utility tray / bedpan holder
19	1	Head traction pulley attachment
20	2	Spring, 1/4 by 1
21	2	Locking pin
22	4	Set screw, 1/4 - 20 by 3/16
23	2	Center pin, 5/8 diameter
24	2	Collar, 5/8 diameter
25	2	Name-plate
26	6	Drive screw
27	2	End turning assembly
28	4	Arm support irons
29	2	Arm support assembly
30	8	Copper rivets
31	1	End swivel, short
32	52	Ever-taut fasteners

3.4 Stability test. When tested as specified in 4.4.1, the orthopedic turning frame shall operate with a minimum of effort, and there shall be no evidence of bending, distortion or other failure on any part of the unit.

3.5 Finish.

3.5.1 Steel rod and tubing. All carbon steel rod and tubing shall be chromium plated over nickel over copper. All corrosion-resisting steel shall be given a No. 4 commercial finish.

3.5.2 Hardware. All screws, bolts, nuts, lock pins, and all other steel parts except steel rods and tubing shall be heavily nickel plated over copper or shall be cadmium plated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

3.5.3 Plywood. All plywood parts shall be sanded smooth, sealed with a wood sealer, and then sprayed with a clear lacquer.

3.5.4 Utility tray. The utility tray shall be given a gray, hammertone, enamel finish.

3.6 Identification marking. Each unit shall have a metal nameplate permanently attached bearing the following information:

Federal stock number.
Name or registered trademark of the manufacturer.
Contract or purchase order number.
Letters U.S.

3.7 Service data. Each frame shall be provided with complete instructions describing fully the proper procedure for preparing the bed for the patient, and for correct handling of the equipment. Instructions shall be silk screened on the underside of the utility tray. Complete instructions shall also be included for unpacking and assembling the unit. In addition, the contractor shall enclose with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts list by the manufacturer's part number. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by the Defense Medical Supply Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Defense Medical Supply Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. Also, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution.

3.8 Workmanship. Workmanship shall be first class throughout. Frame shall be free from defects which detract from its appearance or may impair its serviceability.

3.8.1 Welding. All required joints of the frames shall be continuously welded or joined by brazing. Corners shall be rounded and ground smooth to blend with adjacent areas. Welds shall be durable and sound in every detail. Prior to welding, all components shall be properly aligned in their proper positions to insure a straight, rigid, and level assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:23Z
Check the source to verify that this is the currentversion before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Total (major and minor combined) defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for the stability test shall be in accordance with inspection level S-4 of MIL-STD-105. The acceptance number shall be zero regardless of lot size taken.

4.3 Examination. The frames shall be examined to determine compliance with all requirements contained in this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects *	
Major	
101	Turning frame design not as specified.
102	Moving parts do not move freely.
103	Threads not free of stripping.
104	Means not provided to attach support runners to the cart.
105	Nameplate not supplied, incomplete, or illegible.
106	Metal parts not free of cracks, tears, ruptures, or deformed surfaces.
107	Welds not free of cracks; welds broken or not sound.
108	Service data missing or does not include all specified information.
109	Exposed or accessible edges, corners, or surfaces not free of burrs, sharpness, and sharp projections.
Minor	
201	Plated finish not free of blisters, pits, peeling, or visible porosity.
202	Commercial finish (No. 4) not uniform, not free of scratches, rough grind marks, and other defects.
203	Finish not as specified.
204	Welds not continuous, not clean, not free of pitting, spatter, and discoloration.
205	Exposed welds not ground smooth to blend with adjacent surfaces.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The frames shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:23Z
Check the source to verify that this is the currentversion before use.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Stability test. With the posterior frame in its normal horizontal position, sand bags totaling 300 pounds in weight shall be equally distributed and secured to the frame. After placing the anterior frame over the sand bags, the assembled orthopedic turning frame shall be repeatedly rotated to all possible positions to determine compliance with 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) turning frame, as specified, constitutes one unit.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit shall be disassembled as far as is practicable. Component parts shall be protected by wrapping in kraft paper and cushioning with excelsior pads, paper pads, or flexible corrugated fiberboard. All wrapping and cushioning materials shall be secured by tying with twine or by sealing with kraft gummed paper tape. Component parts shall be packaged in one or more fiberboard cartons or wood boxes of appropriate size and design. Containers shall be adequately secured to prevent accidental opening and shall contain sufficient cushioning material to prevent shifting and breakage.

5.2 Packing.

5.2.1 Level A.

5 2.1.1 Exterior container. Each unit shall be overpacked in an exterior container designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-640, class 2, Grade A, Style E. Closure of the wood boxes shall be in accordance with the appendix of the box specifications. Closure of the fiberboard box shall be in accordance with 30.3.5 of PPP-B-640, except that use of steel strapping is not authorized. Sealing shall be in accordance with 30.2 by the use of tape specified in 30.1.5. Each wood box shall be lined with a waterproof material conforming to UU-P-271.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:23Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Interior packages. Each container shall be marked to identify contents.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6 NOTES

6.1 Intended use. The orthopedic turning frame is intended for mounting on CART, ORTHOPEDIC BED, Stryker.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army. - MD
Navy. - MS
Air Force. - 03

Preparing activity:
Defense Supply Agency - DM

Review Activities
Army. - MD
Navy. - MS
Air Force. - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:23Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:23Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MILITARY SPECIFICATION MIL-T-36310(DM), dated 21 October 1964
TURNING FRAME, ORTHOPEDIC BED, STRYKER, 84 INCH FRAME

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (ATTACH ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS SPECIFICATION. IF THERE ARE ADDITIONAL PAPERS, ATTACH TO FORM AND PLACE BOTH IN AN ENVELOPE ADDRESSED TO PREPARING ACTIVITY)

SUBMITTED BY (PRINTED OR TYPED NAME AND ACTIVITY)	DATE

DD FORM 1426

Source: https://assist.dla.mil -- Downloaded: 20260-21108:232
Check the source to verify that this is the current version before use.

CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

ld Under

DEFENSE SUPPLY AGENCY
EFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

OFFICIAL BUSINESS

POSTAGE FEES PAID
DEFENSE SUPPLY AGENCY

Defense Supply Agency
Defense Medical Supply Center
ATTN: Directorate of Technical Operations (Code T-M5)
3rd Avenue and 29th Street
Brooklyn, N.Y., 11232

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:23Z
Check the source to verify that this is the currenversion before use.